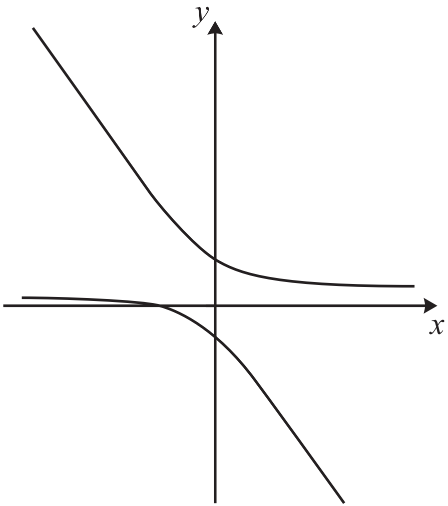
11 The curve in the graph below is defined implicitly by $(2x+y)(y-1)=6$.



(a) Show that $\dfrac{\mathrm{d}y}{\mathrm{d}x}=\dfrac{2(1-y)}{2y+2x-1}$. **[4]**

(b) Find the equation of the normal to the curve at the point (2, 2). **[3]**

(c) Show that there is **no** point on the curve at which the tangent is parallel to the *x*-axis. **[3]**

11(a)

11(b)

(answer space continued on next page)

11(b) **(continued)**

11(c)